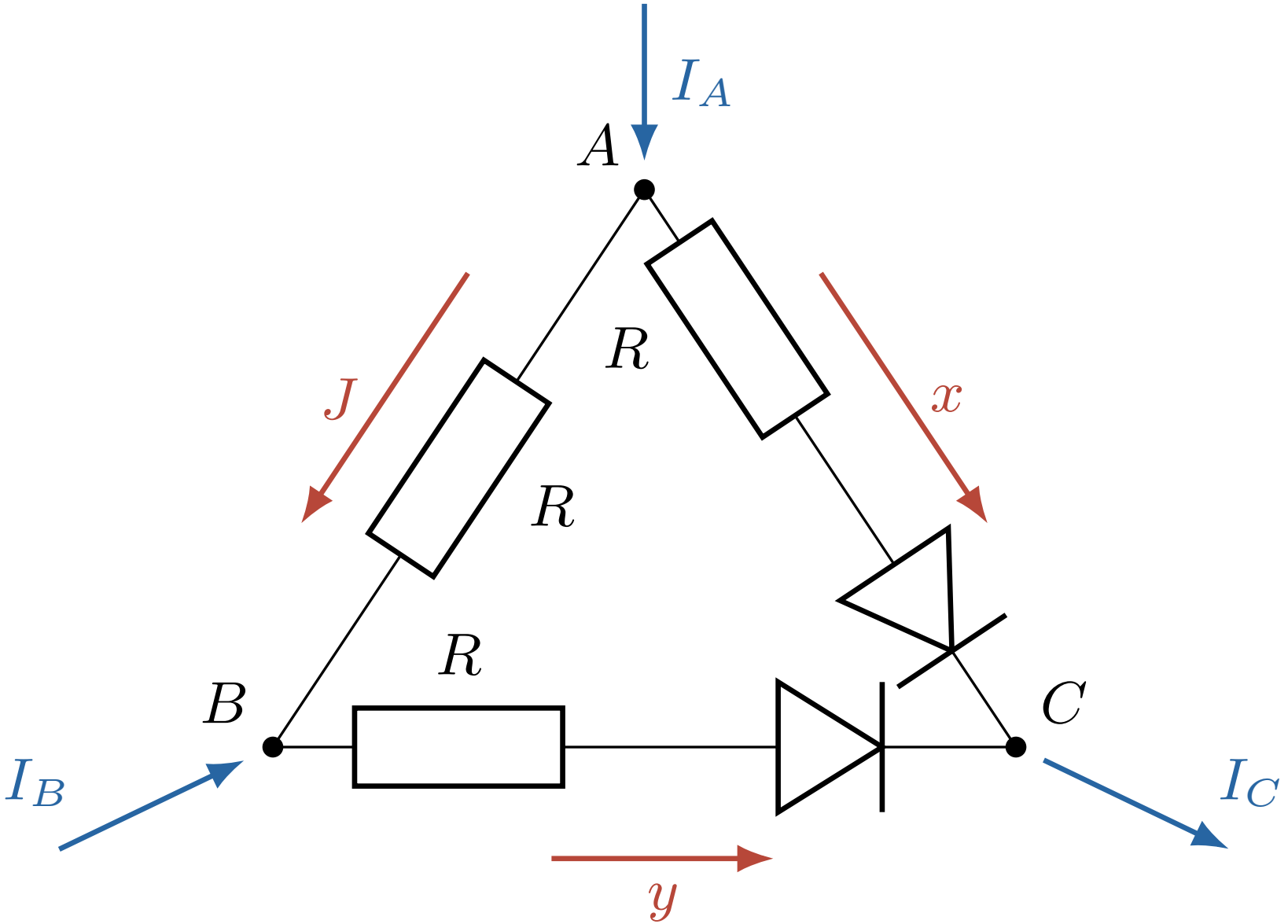

$$I_A = J + x, \quad I_B = y - J$$

$$I_C = x + y = I_A + I_B$$